

Soteria SOW Guardian

Jurisdiction-Aware AI Governance Monitor for KYC Source-of-Wealth Review

Entity: OCBC Bank Ltd | Jurisdictions: Singapore and United States | Option B - Architecture Design

Core thesis

AI can assist source-of-wealth review, but it should not convert uncertainty into false confidence.

Primary buyer

Chief Compliance Officer, Financial Crime Compliance, Model Risk, and AI Governance Committee.

Main output

Green, Amber, Red Friction, and jurisdiction-specific governance gap warnings.

1. Why this product is right for Soteria RegTech

A governance layer, not a replacement KYC system

Company fit

Soteria is good at translating regulatory expectations into controls, evidence checks, audit trails, and jurisdiction-aware workflow logic.

Market wedge

Banks are already deploying AI into KYC workflows. The speed gain creates a governance problem that spreadsheets cannot monitor continuously.

Boundary choice

The product does not sell a simple "compliant" badge. It surfaces uncertainty and forces review.

Bank of Singapore has publicly described an agentic AI Source of Wealth Assistant that reduces report preparation from about 10 days to one hour while retaining human review. This makes the governance problem realistic, not hypothetical. [R2]

2. Problem: polished AI narratives can hide weak evidence

The tool is designed to test evidence, not just document workflow completion

The operational risk

AI can create a coherent source-of-wealth report even when sale proceeds, ownership records, tax documents, or bank transfers do not support the story.

What a memo misses

A memo records that review happened, but not whether each material claim maps to evidence or whether contradictions remain unresolved.

What the tool adds

Claim-to-document mapping, contradiction detection, rule versioning, human challenge monitoring, and model drift alerts.

Design principle: automation can organise evidence, but accountable humans own judgment.

Red Friction blocks straight-through approval when unsupported claims or material contradictions remain.

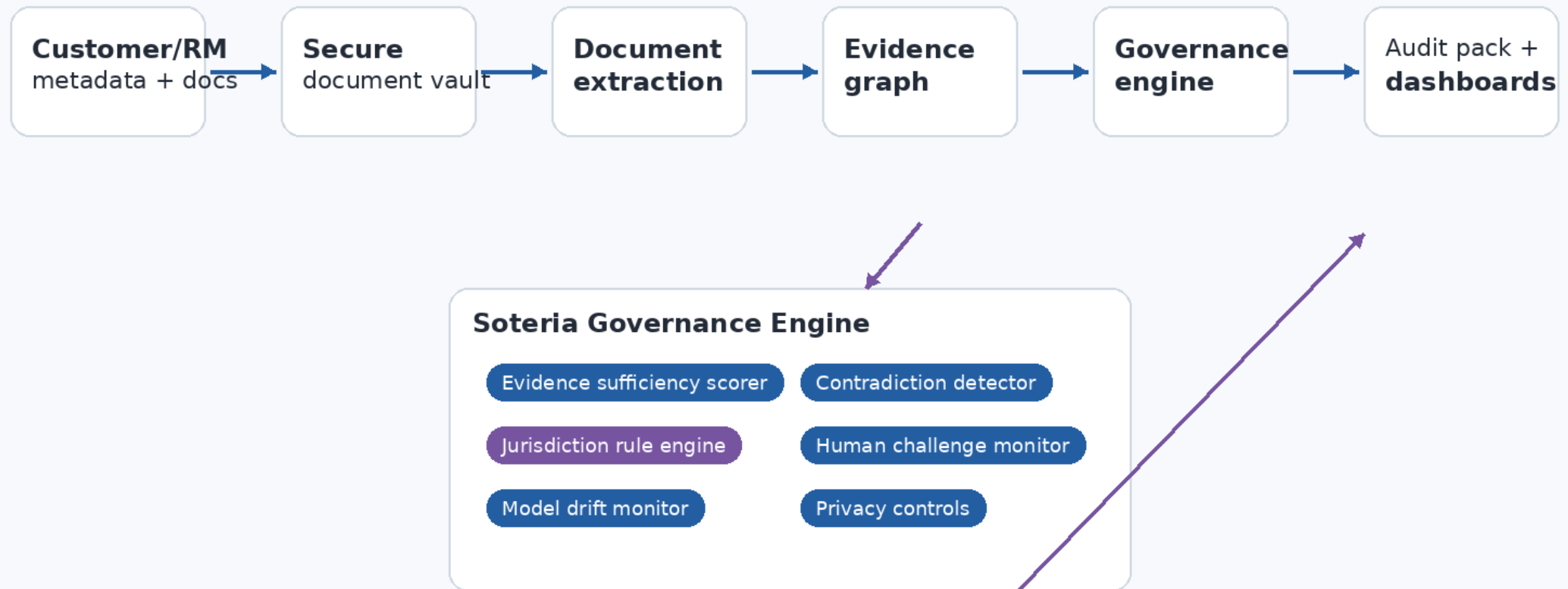
The tool does not reduce escalations as a success metric unless evidence quality improves.

3. Architecture overview

Data flow, evidence scoring, jurisdiction logic, human review, audit pack

Mockup 0 - System Architecture

Data flow, control layer, jurisdiction rule engine, and audit outputs



4. Regulatory divergence handled

Same AI-assisted workflow, different Singapore and US control logic

Issue	Singapore mode	US mode
Agentic AI in KYC	MAS AI MRM + FEAT/Veritas controls	SR 26-2/OCC 2026-13 governance gap warning
SoW evidence incomplete	MAS Notice 626 CDD/EDD review	FinCEN CDD/customer risk profile review
Human rubber-stamping	Accountability weakness flagged	Effective challenge/internal governance weakness flagged
Cross-border case	Dual-rule review	Dual-rule review; no silent downgrade

Jurisdiction awareness is not a country label. It changes rule modules, warnings, owners, and review requirements. [R3-R9]

5. Scoring methodology and synthetic data

The architecture includes simulated inputs and threshold logic

Metric	Formula / meaning	Why it matters
Evidence Coverage Ratio	supported material claims / total material claims	Tests whether the narrative is substantively supported
Unsupported Claim Rate	unsupported AI claims / total AI claims	Detects hallucination and overconfidence
Contradiction Count	material conflicts between AI report and documents	Detects unreliable KYC conclusions
Human Challenge Rate	reviewer amendments/rejections/escalations / AI-assisted cases	Detects rubber-stamping
Model Drift Flag	change in output quality or review corrections	Triggers model-risk review

Synthetic data

30 simulated source-of-wealth cases are included in /data with a data dictionary.

Threshold output

Green, Amber, Red Friction, Amber + Governance Gap, Red Friction + Governance Gap.

Limit

Thresholds are illustrative and not calibrated against real OCBC data.

6. Same case, different jurisdiction output

A synthetic red-friction case shows the rule engine in action

Mockup 2 - Case Jurisdiction Output

Same case, different regulatory meaning in Singapore and the United States

Case SOW-2026-0441

Customer: Titan Pacific Holdings Pte Ltd | Beneficial owner: Mr. L | AI system: Agentic AI Source-of-Wealth Assistant | Case status: RED FRICTION

Singapore mode

- MAS AI MRM controls triggered
- MAS Notice 626 CDD/EDD review active
- FEAT/Veritas accountability checks active
- Evidence coverage below threshold
- Senior compliance approval required

RED FRICTION

US mode

- FinCEN CDD/customer risk profile controls active
- Agentic AI outside specific SR 26-2 scope
- Governance gap warning triggered
- Beneficial ownership review required
- Human approval before acceptance

RED FRICTION + GOVERNANCE GAP

7. Human judgment cannot be outsourced

Human challenge is measured, not assumed

Mockup 4 - Human Challenge Log

Testing whether human-in-the-loop review is meaningful

Reviewer actions for Case SOW-2026-0441

Reviewer: Relationship Manager A

AI recommendation: Accept source-of-wealth narrative

Tool status: RED FRICTION

- ☐ Confirm sale agreement obtained
- ☒ Explain amount mismatch
- ☒ Confirm beneficial owner identity
- ☐ Explain why missing evidence is acceptable
- ☒ Escalate to Financial Crime Compliance
- ☐ Senior compliance review completed

Override rationale

No override allowed yet. The sale agreement is missing and the bank transfer amount does not match the AI narrative. Senior compliance review is required before the source-of-wealth conclusion can be accepted.

Decision: Pending senior review

8. Failure modes and owners

The design names who owns the failure response

Failure mode	Severity	Control	Owner
Rule changes mid-period	High	Versioning, rule-diff alert, open-case impact analysis	Compliance rule owner
Model drift	High	Drift thresholds and model-risk review	Model Risk Manager
Human rubber-stamping	High	Human Challenge Log and reviewer-pattern monitoring	Compliance Head
Jurisdictional misconfiguration	High	Booking-location check, dual-rule activation	Compliance Operations
False negative	Very High	Red Friction + senior compliance approval	Financial Crime Compliance
Privacy failure	Very High	Secure vault, access logs, no uncontrolled model training	Technology Risk / DPO

Failure-mode logic is part of the product, not an afterthought.

9. What the tool does not do - and why

Clarity and honesty for senior management

Boundary	Why this boundary is necessary
No automatic customer approval	Source-of-wealth judgment requires accountable human review
No automatic STR/SAR filing	Reporting decisions require legal/compliance judgment
No simple MAS/US compliant badge	Would hide uncertainty and encourage false confidence
No full transaction monitoring	Scope is AI governance for KYC/source-of-wealth review
No general model training on customer documents	Source-of-wealth data is highly sensitive
No claim that wealth is legitimate	The tool assesses support, contradictions, governance, and auditability

With more time/data, I would calibrate thresholds using real outcomes, validate drift against production AI outputs, test sensitivity to distribution shifts, and integrate with live KYC systems.

10. Senior-management takeaway

Control layer for AI-assisted KYC, not a faster writing tool

Business value

Faster onboarding can be supported without turning source-of-wealth review into a rubber-stamp process.

Regulatory value

Different Singapore and US logic is visible, versioned, and auditable.

Risk value

Weak evidence, contradictions, model drift, and human over-reliance are surfaced early.

Final position: AI may assist compliance work, but it should not convert uncertainty into false confidence.

References [R1]-[R10] are listed in sources/references.md and in the Architecture Design document.